How Scoring Responses Change Attribution Evaluation in a Diffusion Policy

Abstract

Evaluating an explanation of a robot's actions requires a score for changes in its predictions. A robot predicts many action coordinates, so different scores can judge the same explanation differently. This study separates the quantity explained by Integrated Gradients, which uses numerical integration, from the response used to score input removal and restoration. In a diffusion policy, Q is negative squared distance from the reference action; N is negative distance with a small stabilizer. An exact identity shows that, for the same curve, N's normalized area never exceeds Q's when partial input replacement moves predictions no farther than replacing all inputs. Larger changes, called overshoots, can reverse this ordering. In saved simulated cube-picking runs of a 170-million-parameter model with no successful episodes, all case medians of Q-minus-N differences are positive, but extreme calls make most means negative. A matched study of 60 episodes finds that N-target Integrated Gradients exceeds Q-target Integrated Gradients under both responses; both exceed random ordering under N, but neither separates from random under Q. Across three groups, the highest raw mean changes with the response, whereas scaled means and episode medians agree in two groups. Only one of twelve vision comparisons clearly separates the two highest means. Numerical checks on twelve matched contexts of one pretrained model show that 16-bit versus 32-bit calculations and integration rule affect recovery. These samples demonstrate possible numerical failures, not their frequency, and do not verify the original rankings. Response choice, ranking quality and numerical accuracy therefore require separate evidence.

Keywords: integrated gradients, attribution evaluation, diffusion policies, perturbation metrics, numerical integration, robotics, explainable AI

Introduction

Explaining and evaluating robot actions

Attribution methods rank input features, here image patches, language tokens or robot-state coordinates. Testing these explanations can help investigate what influences robot predictions. Insertion starts from a baseline, such as a gray image, and restores highly ranked features first; deletion removes them from the actual input. Fast recovery during insertion and fast change during deletion conventionally favor an explanation[1]. Both tests need a response that reduces prediction change to one score.

Unlike a classifier with a class score, this robot predicts 512 action coordinates per call. Evaluators must choose which output quantity an explanation concerns[2]. Huber and colleagues found different preferred methods for action preference and state value, and instability in endpoint normalization[3]. Zhang and colleagues score vision-language-action interventions with action mean squared error; interpreting it through a Gaussian policy requires assumptions that need not hold for a multimodal diffusion policy[4]. Response choice therefore needs explicit testing.

Diffusion policies refine random noise into action sequences[5]. The Robotics Diffusion Transformer (RDT) conditions predictions on images, instructions and state[6]. Integrated Gradients (IG) multiplies each feature's input change by its average gradient along the baseline-to-input path[7].

The *target* is the quantity explained; the *response* scores predictions after input removal or restoration. For example, Q-target IG explains negative squared action distance while N scores the same predictions by negative stabilized distance. Common practice uses one scalar for both; separating them changes the score without changing rankings or predictions.

What existing evaluation methods establish

Removal-based explanations vary the intervention, model behavior and summary; metrics can follow this structure[8]. Retraining[9], information conveyed by which inputs are hidden[10], and implementation choices[11] affect what scores measure. Insertion/deletion areas under the curve (AUCs) retain feature order but discard attribution magnitude[12]. Meta-evaluation tests metric reliability[13]; rankings need not transfer between datasets[14]. Time-series tests show that low-ranked features can reveal information missed by testing only high-ranked ones[15]. Evaluation thus requires controls and explicit scope.

Input replacement can produce examples outside the training distribution[16]. Ways of hiding inputs change rankings[17]; single-deletion benchmarks address distribution shift and model comparisons[18]. F-Fidelity fine-tunes a replacement model by hiding inputs independently of the explanation and caps evaluation removals at the same budget[19]. Diffusion reconstruction changes masked images[20], while missingness-bias calibration corrects classifier outputs[21]. Here interventions stay fixed: only the scoring response changes.

Output-dependent preferences are already known: TRACE finds different winners for probabilities and unnormalized scores, or logits[22]. Hama, Mase and Owen analyze interactions, response transformations and random-order references[23]; regression baselines also change the question asked[24]. Counterfactual tests challenge visual attribution interpretations[25], StateMask identifies reward-relevant time steps[26], and internal-activation changes can steer vision-language-action behavior[27]. These different behavioral questions motivate specifying the target and response. This study adds an endpoint-normalized identity and tests matched robot contexts.

Completeness means that attributions sum to the baseline-to-input target change, assuming path gradients integrate to that change[28], but does not establish usefulness[29]. Alternative paths change the explanation[30]; IDGI redirects sampled contributions rather than improving quadrature for standard IG[31]. Ground-truth benchmarks[32], synthetic simulations[33] and model-randomization critiques[34] test other properties. Signed contributions may cancel in metrics[35]. Integration-budget checks[36] complement ranking evaluation, but do not establish accuracy for this diffusion policy in 16-bit arithmetic.

The study asks how response choice changes fixed curves, how this appears in successful and failed episodes, whether matched method preferences depend on the criterion, and how precision and integration rule affect recovery. Results follow that order. For saved Q-ranked vision deletion, median AUC is 0.448 under Q versus 0.291 under N: an illustrative, unvalidated 0.40 cutoff judges identical predictions differently.

Methods

Experiments and preserved evidence

An *episode* is one simulated task attempt; a *call* predicts 64 timesteps of seven joint values and one gripper value. A *context* fixes that call's observation, instruction, state and diffusion noise. Two saved PickCube runs each contain thirty episodes and 750 calls, all unsuccessful. PickCube requires moving a cube to a goal. One run ranked by Q, the other by N; both used pretrained RDT-

170M without task fine-tuning[37]. *Rescoring* computes another response from saved intervention predictions, without rerunning the robot or recovering missing observations and noise.

Matched groups comprise pretrained PickCube (60 episodes, 120 contexts), task-trained PickCube (40, 80), and pretrained StackCube (30, 60). They respectively match the saved setting, add a model that often succeeds, and change the task to stacking cubes. The RDT developers' task-trained RDT-1B has about 1.2 billion parameters versus 170 million. Two calls actually used by the simulator are selected uniformly without replacement per episode, under a rule fixed before collection. Numerical tests add all 21 available contexts from the collection underlying twelve completed offline diagnostics, giving 281 contexts. Separate saved runs cover task-trained PickCube, side peg insertion and household-object picking.

Identities are recorded, recovered from unchanged file histories, inferred, or unknown (Supplementary Table S2). Saved code requested the latest model versions rather than recording a revision. Both saved runs' weights are recovered as the same files fixed for the matched study: revision `8aa386cac3bb`, `pytorch_model.bin`, hash prefix `2e87fccee029`; no run-time hash exists. SigLIP image-encoder revision `9fdffc58afc9` and preprocessing (384-by-384 bicubic, scale 1/255, mean/standard deviation 0.5) are recovered; original software versions are mostly unknown. Five denoising steps with the DPM-Solver++ sampler[38] are recorded for Q and recovered for N. Resetting the seed at each call reused one noise draw per episode; missing inputs still prevent exact regeneration.

The matched study preserves model/code versions, observations, state, noise, predictions, rankings, masks and settings. Three machines recorded Python 3.12.3 and Torch 2.8.0+cu128; the fourth runtime is inferred. A fresh machine reproduced 596 attribution maps and 2,466 AUC values for one context bit for bit; two further contexts matched. This verifies those reconstructions, not general hardware independence.

Targets and input replacement

The simulation runs in the ManiSkill3 environment[39]. Our controller executes every fourth predicted action, at most 16 simulator steps per call, and stops an episode by 400 steps. An input *modality* is vision, language or robot state. Holding all other inputs and the noise fixed, let $g(u)$ be the 512-dimensional action prediction when one modality is u. The actual input is x, its chosen baseline is x', and the reference action is $a = g(x)$. Define

$$r(u) = \|a - g(u)\|_2^2, \quad Q(u) = -\frac{r(u)}{2d_a}, \quad N(u) = -\sqrt{r(u) + \epsilon}, \quad d_a = 512, \quad \epsilon = 10^{-12}. \quad (1)$$

Here r is squared action distance. Q scales it by twice the number of action coordinates, resembling a Gaussian log-likelihood without being the policy's likelihood. N uses distance rather than squared distance; its stabilizer makes the gradient finite at zero. The reference action and noise remain fixed throughout each intervention curve.

Vision features are 729 patches in a 27-by-27 external-image grid; five other camera slots stay fixed. A gray-image representation supplies the baseline. Released SigLIP and T5 variants encode images and language, respectively[40,41]. Language features are attended token positions after the adaptor. Saved Q rankings use encoded padding/end-of-sequence tokens; saved N rankings use the RDT developers' precomputed representation with a zeroed baseline. These baselines prevent cross-run language ranking comparisons. State has seven joint readings and one gripper reading, baseline zero. Baselines are not uniquely neutral[42]; interpolated representations may not encode valid language[43].

Let $B = r(x') > 0$ measure the squared action change when every feature is replaced. For either response $S \in \{Q, N\}$,

$$C_S(u) = \frac{S(u) - S(x')}{S(x) - S(x')}, \qquad \Delta = \mathrm{AUC}_Q - \mathrm{AUC}_N. \tag{2}$$

The normalized response is zero at the all-baseline input and one at the actual input. AUC is its trapezoidal area over replacement fractions, using identical weights for Q and N. A *paired* difference scores the same curve two ways. If partial replacement moves the prediction farther than replacing every feature, $r > B$: this is an *overshoot*. Responses and AUCs can then leave the range zero to one and can be very negative when B is small. Normalization is undefined at $B = 0$.

Saved curves use nominal percentages 0, 1, 5, 10, 20, 30, 50, 75 and 100. At an intermediate percentage p, $k = \max(1, \mathrm{round}(Gp/100))$ groups are selected, with half-integer ties rounded to the nearest even integer. Here G counts attended language-token positions. Reconstructed tokenization gives 20, 30, 25 and 16 for PickCube, StackCube, PegInsertionSide and PickSingleYCB. Missing original masks limit run-specific recovery. For $G = 16, 20, 25$, both 1% and 5% select one token: all 3,000 saved PickCube language curves (two runs, 750 calls and two directions) have equal responses at those points. The 800 saved StackCube curves do not. Their composition and realized-fraction checks appear in the supplement. Saved rescoring retains nominal fractions; matched experiments use preserved masks and actual selected fractions.

Integration and matched method comparisons

For target $F \in \{Q, N\}$, IG multiplies each feature's input difference by its average path gradient. With m integration intervals,

$$A_i^F = (x_i - x_i') \int_0^1 f_i(\alpha)\, d\alpha, \qquad E_{i,m} = \frac{x_i - x_i'}{m+1} \sum_{j=0}^{m} f_i(j/m), \tag{3}$$

where $f_i(\alpha) = \partial_i F(x' + \alpha(x - x'))$. The equal-weight estimator E gives the same weight to all $m + 1$ points. The trapezoidal estimate T halves the endpoint weights[44], so exactly

$$E_{i,m} = \frac{mT_{i,m} + (x_i - x_i')[f_i(0) + f_i(1)]/2}{m+1}. \tag{4}$$

The extra endpoint weight can produce error proportional to $1/m$. Exact integration in piecewise-linear networks also exposes sampling error[45]. The saved estimator used 16-bit bfloat16 arithmetic, approximately three significant digits; its 64-interval budget is inferred for the Q-ranked run. It replaced NaN or infinite gradients with zero without counting them. The numerical accuracy of the saved rankings is unverified.

Numerical variants change model precision, summation and rule (Table 1; Supplementary Table S8). Corrected 16-bit calculations sum at 32 bits, about seven significant digits. Midpoint and Gauss-Legendre rules use interior nodes; adaptive Gauss-Kronrod adds nodes using an error estimate. Without an exact policy answer, qualification requires adaptive convergence and pairwise agreement among adaptive integration, 128-node Gauss-Legendre and the 512-interval 32-bit trapezoid: relative L1 difference below 0.1%. Relative L1 is the absolute coordinate-difference sum divided by the absolute comparator-coordinate sum. Supplementary Table S5 separates the diagnostic populations.

Even a linear action example exposes endpoint sensitivity. For $g(u) = u$, baseline zero and input one, exact N attribution is 0.999999, slightly below one because of the stabilizer. At 64 intervals,

Table 1 | Study groups answer different questions. Counts are episodes (successes) / calls unless stated. Precision is model / accumulation: bf16 is 16-bit, fp32 32-bit. Budgets count intervals; equal weighting uses one extra evaluation. Asterisks indicate inferred budgets. The saved 1B rerun is excluded. Preserved and offline samples overlap. Saved runs measure response shifts, matched groups compare methods, and other contexts test numerics. Saved 170M weights are recovered; matched weights are recorded. Tables S2 and S8 give full identities, statuses and variants.

Group	Episodes / calls	Rule and budget	Precision	Preserved
Saved PickCube 170M, target Q	30 (0) / 750	Equal weights, 64*	bf16 / bf16	Scored curves
Saved PickCube 170M, target N	30 (0) / 750	Equal weights, 64	bf16 / bf16	Scored curves
Saved PickCube 1B, target Q	60 (15) / 1,199	Equal weights, 64	bf16 / bf16	Scored curves
Saved peg and object tasks, 170M, target Q	16 (0) / 400 each	Equal weights, 128*	bf16 / bf16	Scored curves
New PickCube 170M, targets Q/N	60 (0) / 120	Trapezoid, 512; checks	bf16 / fp32; fp32 subset	Complete contexts
New PickCube 1B, targets Q/N	40 (31) / 80	Trapezoid, 512; old rule	bf16 / fp32	Complete contexts
New StackCube 170M, targets Q/N	30 (0) / 60	Trapezoid, 512; old rule	bf16 / fp32	Complete contexts
Preserved four-task sample	21 contexts	Numerical checks	bf16 and fp32	Contexts
Separate offline diagnostics	6 / 12 per input and target	Trapezoid, 128 to 1,024	fp32 / fp32	Contexts and outputs

equal endpoint weights underestimate it by 1.54%, and trapezoidal weights by 0.78%. Midpoint and 64-node Gauss-Legendre estimates slightly exceed it, each by about one part per million. N's gradient falls sharply to zero near the actual-input endpoint. Over the tested budgets, that unresolved transition gives approximately first-order trapezoidal error. With fixed positive epsilon the integrand is smooth and this is not a universal asymptotic claim.

Seven methods rank groups by summed absolute values: Q/N IG, Q/N mean path gradients, input difference, and Q/N original equal-weight IG. Path gradients omit IG's input-difference factor; input difference ranks baseline-to-input change alone. Ordinary IG/path gradients use 512 trapezoidal intervals, a 16-bit model and 32-bit accumulation. Chance controls average 32 image orders, 64 token orders and all $8! = 40{,}320$ state orders. Raster, reverse and center-out image orders, plus forward/reverse token or joint orders, test attribution-free alternatives.

Advantage is method-minus-random AUC for insertion and random-minus-method for deletion, so larger values favor a method. The primary criterion averages within each episode, then

across episodes having defined values for every method. Subtracting a common random reference cannot reorder methods within one response. Scaling each context's advantage by its random-order standard deviation before episode averaging changes the weighting and can change the winner; the state reference enumerates every order. Episode medians and illustrative deletion cutoffs of 0.40 and 0.50 provide separate sensitivity checks.

The collection protocol, analysis plan and three amendments were recorded before their governed outcomes were examined. The seven primary pretrained-PickCube vision-deletion contrasts in Table 2A use 10,000 bootstrap draws that resample entire episodes to retain dependence between their calls. Bonferroni adjustment divides the error allowance among seven comparisons, giving alpha $0.05/7$. Best-versus-second intervals use $0.05/4$ for each group's four cells, two responses times two directions. Equal highest values have no unique winner. These selected comparisons are exploratory and do not jointly test a winner across all groups. Simulated interval coverage was checked before examining results. Saved-run medians and positive shares were emphasized retrospectively because means are tail-sensitive, but all means remain reported. Intervals describe recorded episodes, not variation across training runs, checkpoints or tasks.

A separate retrospective report uses 10,000 paired episode-bootstrap draws, seed 930011, to count matching winners, different winners and ties under raw means, scaled means and episode medians. Success/failure median intervals resample whole episodes within each outcome. These describe observed-sample stability, not probabilities of a true population reversal.

Completeness residual is the absolute attribution-sum error divided by the absolute target change. Coordinate differences, Spearman rank correlation, the share of top-ranked groups in common, overlap of replaced-feature sets and AUC changes are reported separately. Zero denominators remain undefined. The 1% completeness and 0.01 AUC-change thresholds are study-specific diagnostics, not guarantees. Separate 32-bit diagnostics cover twelve contexts per modality/target from six episodes across four tasks, with no completed RDT-1B context. They demonstrate possible numerical failures without estimating frequency.

All experiments are simulated, involve no human or animal subjects and require no IRB approval.

Figure 1 connects the workflow, scoring responses and constructed examples.

Results

Response identity and saved curves

Exact self-reference means that the reference action is the model's own actual-input prediction, giving $r(x) = 0$. The saved curves satisfy this to floating-point precision. Equation 2 yields

$$C_Q - C_N = \frac{(\sqrt{r+\epsilon} - \sqrt{\epsilon})(\sqrt{B+\epsilon} - \sqrt{r+\epsilon})}{B}. \tag{5}$$

The difference is positive when $0 < r < B$, zero at $r = 0, B$, and negative at overshoots. If every sampled point is in range, $r \leq B$, then $\Delta \geq 0$, strictly when some positive-weight point satisfies $0 < r < B$. Thus N makes the same ranking look better for deletion and worse for insertion. The saved schedule gives $\Delta \leq 0.2175$ for every defined curve. Figure 1B–D illustrates the score, identity and a constructed target reversal; proofs are in Supplementary Equations S1–S6.

The usual 0.5 benchmark corresponds to a straight-line normalized response; random-order areas need not equal it. If every feature contributes equally and independently, every ordering gives insertion/deletion AUC 0.65976 under Q on the saved 100-feature schedule, versus 0.5 under un-

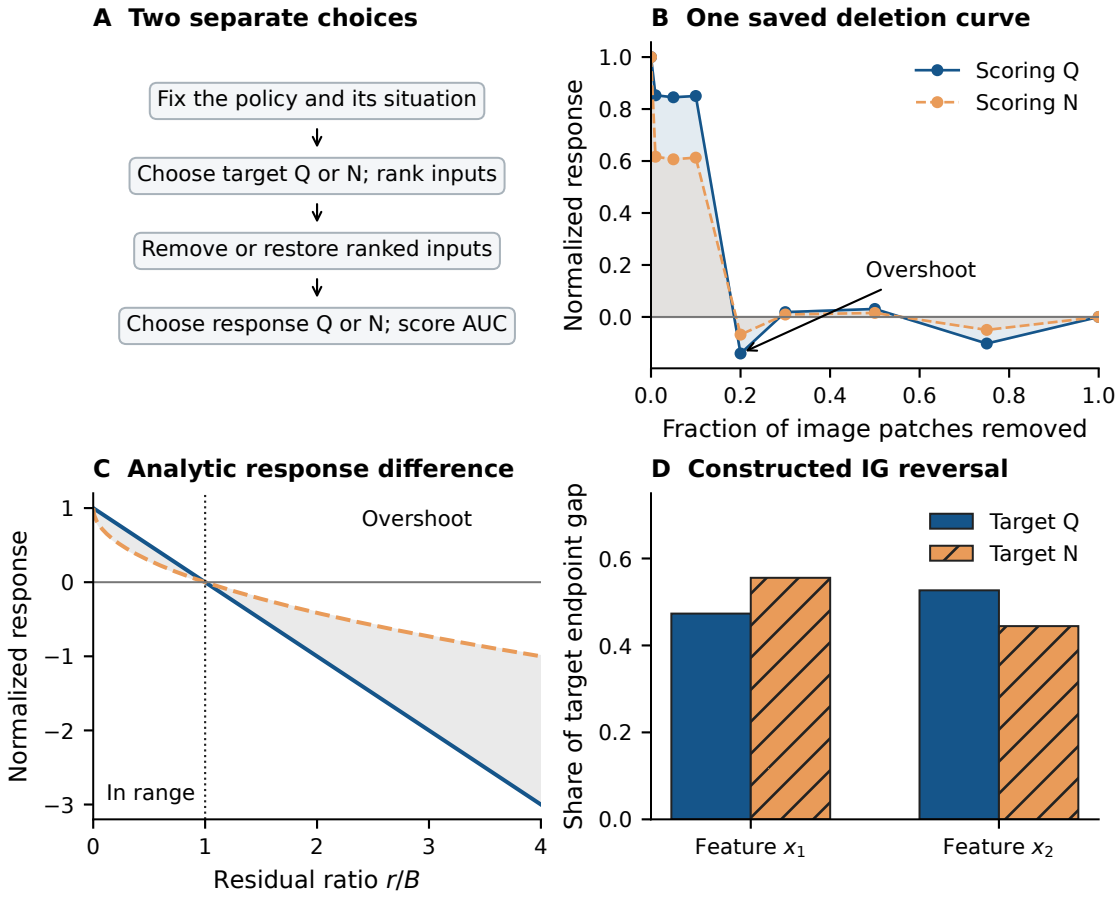


Figure 1 | Target and scoring response are separate choices. A) Workflow. **B)** One actual saved curve scored by Q and N; shading defines AUC and $r > B$ marks an overshoot. **C)** Analytic responses versus squared residual ratio r/B, with their gap in the zero-stabilizer limit. **D)** IG shares for $g(x_1, x_2) = x_1^2 + 0.8x_2$, divided by each target's total change. Labels identify input features. The constructed reversal is not an observed PickCube result; the supplement gives the function and proof.

stabilized N. Thus 0.5 misstates Q's chance performance. These are analytic examples, not RDT controls. A three-feature construction also reverses ordering preference by changing only the response (supplement).

All eight saved-case medians are positive: two ranking targets, two modalities and two directions. Medians range from 0.1041 to 0.1590; 75.6%–94.3% of calls have $\Delta > 0$, and 16.0%–74.0% overshoot. These medians measure the size of a response shift whose fully in-range direction is predicted; positivity of a median in an overshooting population remains empirical. Six means are negative, and three corresponding intervals span zero. For Q-ranked vision insertion, the mean is -17.1: the 30 first calls contribute -16.92 and the remaining 720 contribute -0.18, both divided by all 750 calls. The same seven of thirty episodes supply differences below -100 in both saved runs, so their tails are not independent confirmations.

Figure 2B shows 29–144 of 750 calls per case below -0.05, none above 0.25: positive centers hide substantial tails. Supplementary Table S1 gives full summaries. Shared simulator seeds do not match contexts: Pearson correlation of log B across thirty aligned episodes falls from 0.992 at call 0

to 0.289 at call 2, then ranges around 0.4–0.8 and ends at 0.66. Head-to-head ranking comparisons are therefore invalid. N-ranked vision B has fifth/ninety-fifth percentiles 2.9/80.8.

Small B magnifies intermediate prediction changes. Across 10,796 saved case calls, 335 have $\Delta < -1$ and 23 have $\Delta < -100$. Equation 5 itself requires the maximum interior residual ratio to exceed approximately 2.83 and 126.2, respectively; these thresholds are not independent evidence for a mechanism. Observed minimum ratios are 3.83 and 252.8, and the most extreme calls have $B \leq 0.514$. Concentration in first calls and correlations from -0.56 to -0.98 across sixteen overshooting groups support a possible small-denominator mechanism. Whether partially replaced inputs overshoot because they resemble nothing in training remains untested. Supplementary Figure S1 shows the detailed geometry.

Successful episodes and additional tasks

Saved task-trained PickCube has fifteen successful episodes (74 calls) and forty-five failed episodes (1,125 calls) across three seeds. Successes stop early, emphasizing early calls. Successful/failed vision medians are 0.0653/0.0868 for insertion and 0.0747/0.1035 for deletion. Overshoot rates are 24.3%/29.0% and 41.9%/31.6%, respectively. More extreme failure minima partly reflect fifteen times as many calls. Supplementary Table S6 gives rates and intervals, keeping language separate by seed because baselines differ.

Task-trained PickCube succeeds in 31/40 episodes (62 successful, 18 failed calls); both pretrained groups have no successes. Q-target trapezoidal IG gives successful/failed medians 0.0616/0.0913 for insertion and 0.0212/-0.0086 for deletion. The failed deletion interval $[-0.1775, 0.0608]$ rests on nine episodes; its sign is not interpreted. Deletion overshoots affect 41/62 successful and 13/18 failed calls; insertion counts are 13/62 and 7/18. The largest extremes occur in successes, opposite the saved pattern, but successes have more observations. Different estimators, contexts and runtimes preclude causal success claims.

The saved peg-insertion and household-object tasks each contain sixteen unsuccessful episodes and 400 calls, using a 128-interval estimator. Their vision insertion/deletion medians are 0.0990/0.1152 and 0.0954/0.1405. Peg insertion retains a large negative insertion tail: mean -3.54, minimum about -212, and overshoot share 69.0%. Figure 2 compares these separate populations. All successful evidence in this study comes from PickCube.

Matched rankings and criterion sensitivity

On sixty pretrained-PickCube episodes, N-target IG exceeds Q-target IG under both responses (Table 2A). Both exceed random ordering under N; neither separates from random under Q, whose tails widen intervals. Changing both target and response gives -0.438, adjusted interval $[-3.91, 2.44]$, an inconclusive comparison in different normalized units.

Raw-mean vision-deletion winners differ between Q and N in all three groups; scaled means and episode medians agree in both PickCube groups but differ in StackCube (Table 2B). Only one of twelve vision cells separates its top two means: pretrained PickCube Q deletion, 0.0998, adjusted interval $[0.0181, 0.315]$. Two language cells also separate, giving three of twenty overall. Supplementary Table S3 reports every modality and selection flag.

Fixed orders beat every attribution method's raw mean in three vision cells: pretrained PickCube insertion under Q and N, and task-trained PickCube deletion under Q. The latter winner, input difference, also falls below random. Scaling and medians change these comparisons (Supplementary

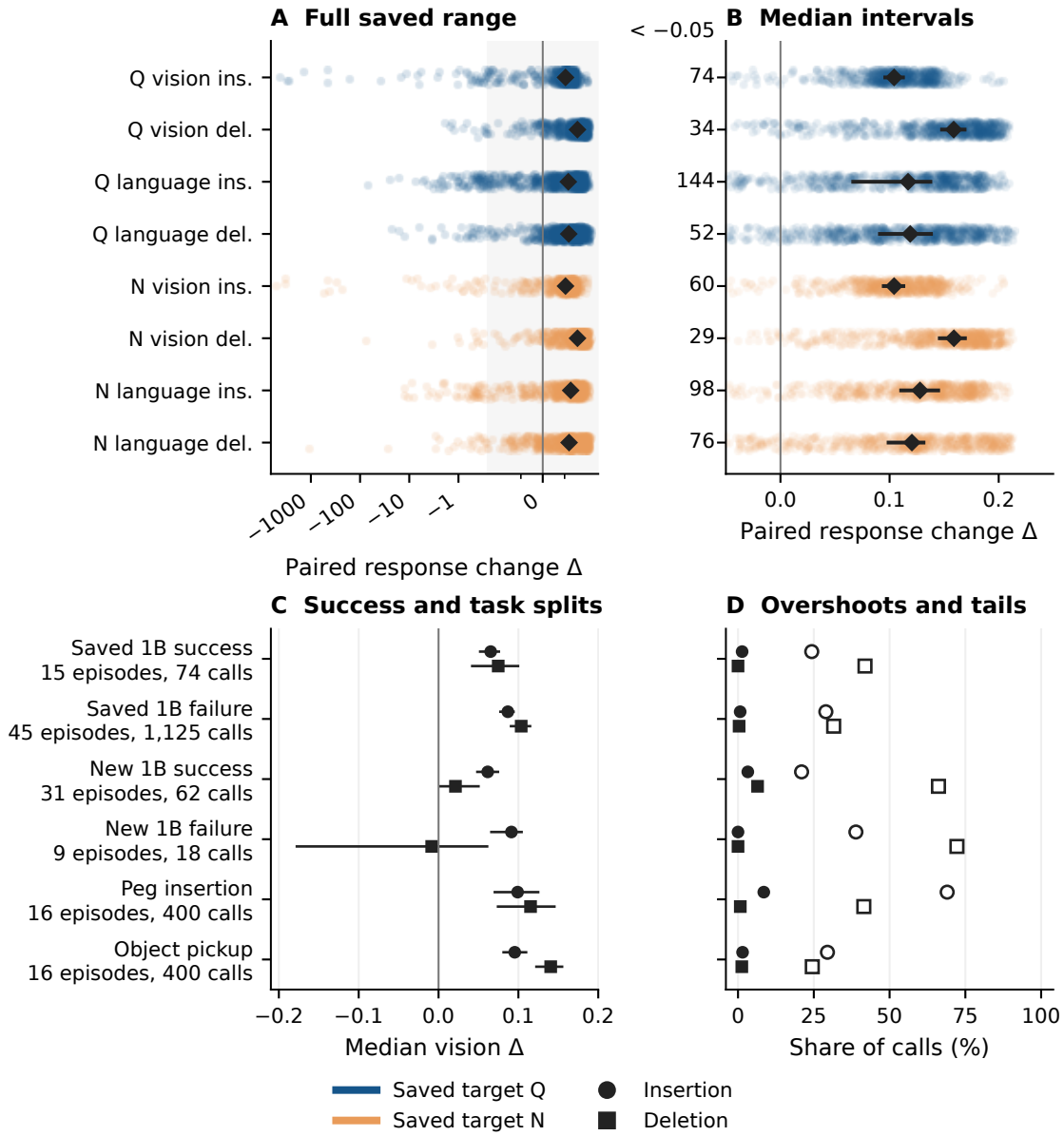


Figure 2 | Positive centers coexist with tails in successful and failed episodes. A) All eight saved distributions, 750 calls each. Symmetric-log scaling is linear for $|\Delta| \leq 0.25$, compressing larger magnitudes without losing sign. **B)** Central medians, 95% episode-bootstrap intervals and left-edge counts below -0.05. Q/N row labels denote ranking targets; $\Delta = \mathrm{AUC}_Q - \mathrm{AUC}_N$. **C)** Vision medians and intervals for separate success/failure and additional-task populations. **D)** Overshoot shares (open) and $\Delta < -1$ shares (filled); circles: insertion, squares: deletion. Table 1 identifies estimators; Supplementary Tables S1 and S6 give full values and seed membership. Repeated calls are dependent.

Table 2 | Primary contrasts and response-dependent choices. A: seven preplanned vision-deletion contrasts on 60 PickCube 170M episodes, with Bonferroni-adjusted intervals. The last row mixes response units. B: vision-deletion point-estimate winners. Scaling divides each call's advantage by its random-order standard deviation before averaging; medians use episode means. IG uses a 512-interval trapezoid, equal-weight IG 64 intervals, both with 16-bit model calculations. Winners need not separate statistically. Supplementary Tables S3 and S4 retain all comparisons and controls.

A. Primary contrasts

Contrast	Q: estimate [interval]	N: estimate [interval]
IG-Q minus random	0.0280 [−3.54, 3.12]	0.419 [0.327, 0.556]
IG-N minus random	2.89 [−3.41, 14.6]	0.466 [0.341, 0.704]
IG-Q minus IG-N	−2.86 [−11.4, −0.0115]	−0.0467 [−0.146, −0.00846]
IG-Q at Q minus IG-N at N	−0.438 [−3.91, 2.44]	Mixed responses

B. Point-estimate choices

Group (episodes)	Score	Raw mean	Scaled mean	Episode median
PickCube 170M (60)	Q	Path-gradient N	IG-N	IG-N
	N	Equal-weight IG-Q	IG-N	IG-N
PickCube 1B (40)	Q	Input difference	Path-gradient N	Path-gradient N
	N	Path-gradient N	Path-gradient N	Path-gradient N
StackCube 170M (30)	Q	Path-gradient N	Path-gradient N	Path-gradient N
	N	Equal-weight IG-N	Equal-weight IG-N	IG-N

Table S4). Unvalidated 0.40 and 0.50 cutoffs each disagree with calibrated preferences in four of six deletion cells. Supplementary Table S7 reports bootstrap winner stability with ties separate.

Numerical recovery and diagnostic limits

Most saved vision rankings miss the study's 1% completeness criterion: median residuals are 1.96% for Q-ranked and 4.76% for N-ranked calls, with 77.2% and 99.6% above 1%. Their sums fall short of the baseline-to-input target change in 87.5% and 99.1%. This one-sided deficit fits endpoint bias; storage rounding alone can err in either direction. Individual coordinate errors remain unknown. Figure 3 compares matched numerical recovery.

On twelve matched pretrained-PickCube vision contexts, increasing trapezoidal intervals from 64 to 512 changes 16-bit median completeness residuals from 4.76% to 4.46% for N and 1.05% to 0.991% for Q. With 32-bit model calculations, N falls from 1.54% to 0.192% and Q from 0.0261% to 0.000321%. Additional 16-bit points help little. At sixteen intervals, 32-bit midpoint N reaches about 0.0275%. Rule and interval count both matter.

On 120 new pretrained-PickCube vision contexts, original-estimator median relative L1 differences from the corrected 16-bit trapezoid are 2.07% for N and 1.82% for Q. The 32-bit subset has twelve contexts; only eight have stored 16-bit rule-comparison maps (Supplementary Table S5). Rank correlations stay at least 0.9993; median top-5% overlap is one. Yet N-target deletion AUC

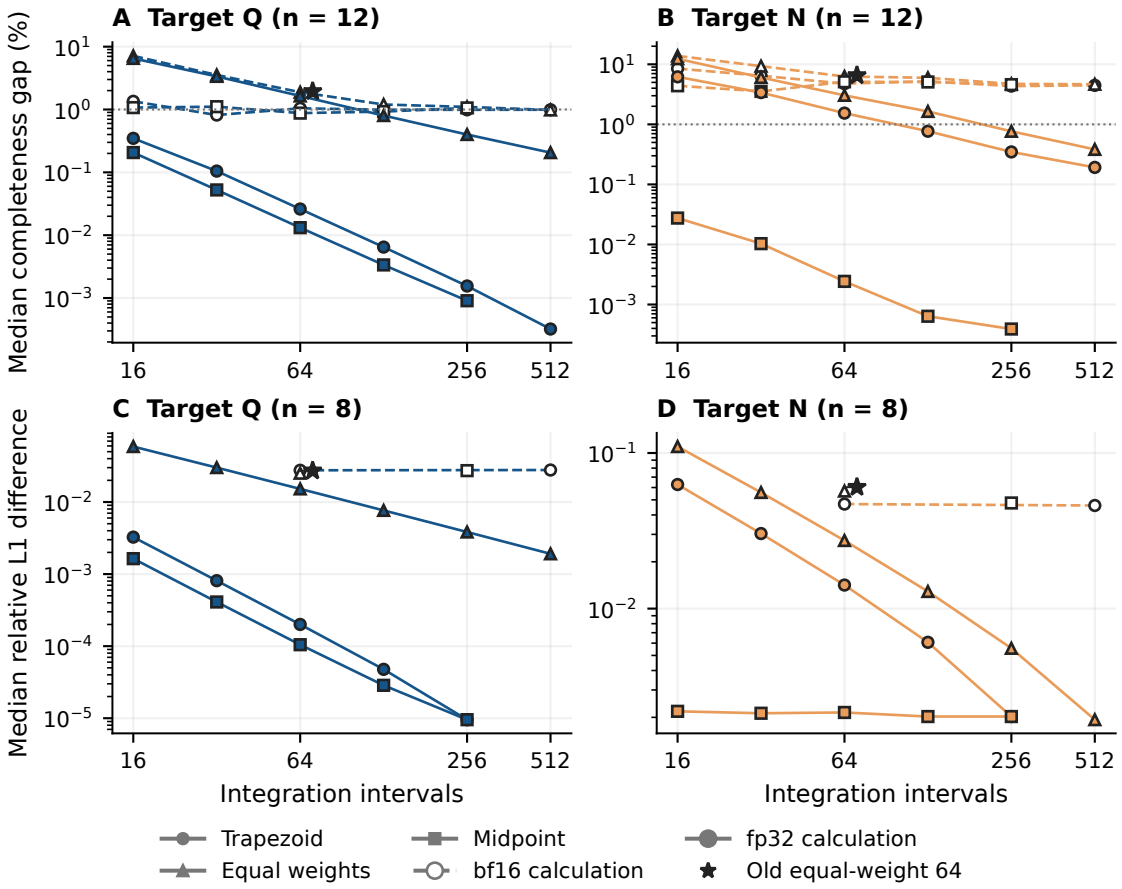


Figure 3 | Precision and rule affect recovery on matched contexts. A)–B) Median completeness residuals, twelve pretrained-PickCube vision contexts; horizontal line: study-specific 1% threshold. **C)–D)** Relative L1 differences from the 32-bit, 512-interval trapezoid, eight stored-map contexts. Open/dashed markers/lines indicate 16-bit calculations, filled/solid 32-bit; shapes identify rules. Original-estimator stars are offset for visibility. The comparator is not exact IG: its N endpoint discrepancy contributes to D's midpoint plateau. Its zero self-difference is omitted on logarithmic axes.

changes exceed 0.01 in 7/120 N-scored and 12/120 Q-scored contexts; Q-target counts are 4/120 and 12/120. The latter median/max are 0.0018/221: normalized AUC is unbounded below, so stable rankings need not give stable scores.

All 815,210 evaluated nodes are finite; no 64-bit model run was performed. Adaptive integration converges in all 48 comparisons, but three-way qualification passes only 15/24 Q and 5/24 N cases: eight contexts (four new, four stored), three modalities and two targets. For N vision/language, adaptive and Gauss-Legendre estimates agree while the trapezoid differs by about 0.17%; state also differs between the first two. Endpoint effects are plausible, but qualification remains failed and agreement does not prove accuracy.

The separate four-task sample shows agreement of repeated function evaluations despite attribution changes with integration budget (Supplementary Figure S4; Table S5). It demonstrates possible numerical failures, not their frequency or the saved rankings' correctness. The supplement's

smooth constructed example passes completeness and grid-agreement checks while reversing exact coordinate order. This is not an observed policy-ranking reversal.

Discussion

Overshoots produce tails in both successful and failed runs. The primary matched experiment favors N-target IG; exploratory winners depend on averaging and scaling, with eleven of twelve vision top-two intervals including zero. Subtracting a common random reference cannot reorder methods. Raw-mean preference is therefore not general superiority.

The findings extend Huber's and TRACE's output-dependent preferences. Hama considers monotone transformations[23]; N is strictly increasing in Q. Its condition is a strictly increasing transformation of additive feature contributions on binary inputs. That can hold here since $x_1^2 = x_1$ for binary inputs. Its IG argument uses constant feature slopes before transformation, whereas our quadratic construction changes gradient direction. This different setting is not a counterexample. Our additions concern normalization, overshoots, separate targets/responses and matched numerical comparisons.

Like TRACE and Blücher, these findings show metric sensitivity. They extend Makino's integration-budget tests to precision and rule choice. Zhu reconstructs interventions; here predictions stay fixed.

Q's tails leave IG-versus-random comparisons inconclusive. Nine failed episodes give an uncertain negative median, while larger extremes occur among successes; neither implies causation. Numerical qualification remains failed despite plausible endpoint effects, and most saved rankings miss the completeness criterion. The 0.448/0.291 example illustrates sensitivity to an illustrative 0.40 rule, not justification for it.

State targets and responses because they answer different questions; preserve references, noise, baselines and groups to repeat comparisons. Random and fixed controls reveal what scores reward. Means, medians, tails and undefined cases expose different sample properties; resampling episodes retains call dependence. Completeness, coordinate, rank and score agreement need separate checks because none guarantees the others.

Limitations and conclusion

The saved rankings' numerical accuracy remains unverified; missing inputs prevent regeneration and their Q/N runs are unmatched. Original pretrained PickCube evidence is entirely unsuccessful; all successful evidence comes from task-trained PickCube. Only one model family, simulation, two responses, chosen baselines/groups and mostly gradient-based methods are studied. No weight-randomization test of dependence on learned weights was performed[46]. The tail mechanism, including distribution shift from partial replacement, remains untested. Small numerical samples show possible failures, not frequencies. Other successful policies, responses, attribution families and physical robots require matched studies.

The identity proves an in-range ordering for fixed predictions. Unsuccessful simulated PickCube runs show its size and tail-dominated means; successful runs also overshoot. Primary comparisons favor N-target IG. Raw-mean winners change in all three groups; scaled means and episode medians differ only in StackCube. Numerical experiments show possible failures, not their frequency, and do not validate original rankings. Evaluation must establish scoring choice, numerical recovery and ranking quality separately.

Data and code availability

A CPU reproduces saved response summaries, plots, success splits and completeness checks. Missing observations and noise prevent original execution and ranking regeneration. Matched inputs support verified fresh-machine reconstruction. Packages supply code, plans, records, identities, supplementary results and replay instructions. Anonymous materials are available through the review archive. This unlisted link requires no account; anyone holding it can download. The archive distinguishes current manuscript materials from scientific replay evidence.

AI-use disclosure

OpenAI Codex, Claude Code, pi.dev and Cursor assisted with code, experimental planning/execution, analysis, statistics, literature searches and reference auditing, figure-generation code, and number/claim verification. AI tools drafted and revised the manuscript, captions, tables, supplementary derivations, response letter and reviewer documentation, including archive/regeneration guides. Earlier work tested StealthGPT on a discarded version and used Pangram to assess an earlier draft. The author retains responsibility; the prose was not composed without AI assistance.

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